# *Remarks Concerning Spin Manifolds*

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This paper will consist of several unrelated remarks. In §2, as an application of the concept of "spin manifold," a non-standard involution of the 7-sphere is constructed. In §3 the spin cobordism ring is discussed.

## 1. Spin structures

Let $\xi$ denote an oriented vector bundle with base space $B$. The concept of a "spin structure" on $\xi$ can be defined in various ways. (See [4] §6.5, and [11].) The following definition is perhaps easiest to use. Let $P\xi$ denote the total space of the associated principal bundle. (One may use either the principal bundle with the rotation group $SO(n)$ as fibre, or with the linear group $GL^+(n, R)$ as fibre. It will not matter which is used, since the first is a deformation retract of the second.) First suppose that $n \geq 2$.

DEFINITION. A *spin structure* on $\xi$ is a cohomology class $s \in H^1(P\xi, Z_2)$ whose restriction to each fibre is non-zero.

For the special case $n = 1$ we will allow $s$ to be an arbitrary element of $H^1(P\xi, Z_2) \cong H^1(B; Z_2)$.

Such a class $s$ determines a 2-fold covering of $P\xi$. This covering can be taken as the total space of a new principal bundle whose structural group is the 2-fold covering group of $SO(n)$ or $GL^+(n, R)$.

The following three lemmas give the basic properties of this concept.

LEMMA 1. *$\xi$ admits a spin structure if and only if the Stiefel–Whitney class $w_2(\xi)$ is zero. This spin structure is unique if and only if $H^1(B; Z_2) = 0$.*

LEMMA 2. *Let $U_1$ and $U_2$ be open sets whose union is the base space $B$. Given spin structures $s_1$ on $\xi|U_1$ and $s_2$ on $\xi|U_2$ which coincide on $\xi|U_1 \cap U_2$, there exists a spin structure $s$ on $\xi$ which extends both $s_1$ and $s_2$.*

(Caution: In general $s$ will not be uniquely determined by $s_1$ and $s_2$.)

LEMMA 3. *Given spin structures on two of the three bundles $\xi$, $\eta$, $\xi \oplus \eta$, there is a uniquely determined spin structure on the third.*

PROOFS. Lemma 1 follows (as in [11]) from the exact sequence

$$0 \to H^1(B) \to H^1(P\xi) \to H^1(\text{Fibre}) \to H^2(B),$$

where the coefficient group $Z_2$ is understood, and $B$ is assumed to be connected. Lemma 2 follows from the Meyer–Vietoris sequence

$$\tilde{H}^0(P_1 \cap P_2) \to H^1(P\xi) \to H^1(P_1) \oplus H^1(P_2) \to H^1(P_1 \cap P_2),$$

where $P_i$ stands for $P(\xi | U_i)$.

To prove Lemma 3, first consider the Cartesian product $\xi \times \eta$, and note that $P\xi \times P\eta \subset P(\xi \times \eta)$. Given spin structures $s \in H^1(P\xi)$ and $t \in H^1(P\eta)$, consider the class $s \times 1 + 1 \times t \in H^1(P\xi \times_d \eta)$.

ASSERTION. There exists one and only one cohomology class $u \in H^1(P(\xi \times \eta))$ which is a spin structure, and which extends the class $s \times 1 + 1 \times t$.

If $B$ is connected then this fact is proved by chasing around the following diagram.

$$\begin{array}{ccccccc} & H^1(B \times B) & \to & H^1P(\xi \times \eta) & \to & H^1(\text{Fibre}) & \to H^2(B \times B) \\ & \downarrow & & \downarrow & & \downarrow & \downarrow \\ 0 \to & H^1(B \times B) & \to & H^1(P\xi \times P\eta) & \to & H^1(\text{Fibre} \times \text{Fibre}) & \to H^2(B \times B) \end{array}$$

The general case follows easily.

Now let $\Delta\colon P(\xi \oplus \eta) \to P(\xi \times \eta)$ denote the diagonal map. Then we can form the cohomology class $\Delta^* u \in H^1P(\xi \oplus \eta)$.

DEFINITION. $\Delta^* u$ is the spin structure on $\xi \oplus \eta$ which is determined by $s$ and $t$.

Finally we must verify that each of the cohomology classes $s$, $t$, $\Delta^* u$ is uniquely determined by the other two. The most general spin structure on $\xi$ is $s + \pi_\xi^* a$, where $\pi_\xi\colon P\xi \to B$ denotes the projection map and $a$ denotes an arbitrary element of $H^1B$. If we replace $s$ by $s + \pi_\xi^* a$ and similarly replace $t$ by $t + \pi_\eta^* b$, then $\Delta^* u$ will be replaced by $\Delta^* u + \pi_{\xi \oplus \eta}^*(a + b)$. It is now clear that any two determine the third uniquely. This completes the proof of Lemma 3.

As an application of Lemma 3, consider a smoothly imbedded, oriented manifold $M^n \subset R^{n+k}$, with tangent bundle $\tau$ and normal bundle $\nu$. Then $\tau \oplus \nu \cong \tau' | M^n$, where the tangent bundle $\tau'$ of $R^{n+k}$ has a unique spin structure. Thus a spin structure on $\tau$ determines a spin structure on $\nu$ and conversely.

DEFINITION. The pair consisting of a manifold $M^n$ together with a spin structure on its tangent bundle is called a *spin manifold*.

As another application of Lemma 3 one sees that the boundary of any spin manifold is again a spin manifold. Similarly the product of two spin manifolds is a spin manifold.

## 2. An exotic involution of $S^7$

In order to illustrate the utility of the concept of spin structure, we will use it to prove the following.

THEOREM 1. *There exists a smooth involution $f: S^7 \to S^7$ without fixed points so that the orbit manifold $S^7/f$ is not diffeomorphic to $P^7$.*

The proof will depend on an invariant which Eells and Kuiper [6] have defined for certain spin manifolds of dimension $4k - 1$. In order to simplify the exposition we will define this invariant only for 7-manifolds.

First consider a closed 8-dimensional spin manifold $M$ with signature $\sigma$. According to Thom and Hirzebruch [7] one has the formula

$$\sigma = \tfrac{1}{45}(7p_2 - p_1^2)[M].$$

On the other hand Atiyah and Hirzebruch [3] define the $\hat{A}$-genus

$$\hat{A}[M] = \tfrac{1}{5760}(7p_1^2 - 4p_2)[M]$$

and prove that this is also an integer (for a spin manifold). Eliminating $p_2[M]$ between these two linear equations, we obtain the formula

$$\hat{A}[M] = (p_1^2[M] - 4\sigma)/896.$$

Now consider a compact 8-dimensional spin manifold $(W, s')$ with boundary $(\partial W, s)$. Following Eells and Kuiper, one must assume:

HYPOTHESIS. The natural homomorphism $H^4(W, \partial W; Q) \to H^4(W; Q)$ between rational cohomology groups is an isomorphism.

The signature $\sigma$ of $W$ is easily defined. Furthermore, after lifting the Pontrjagin class $p_1$ of $W$ back to $H^4(W, \partial W; Q)$, the Pontrjagin number $p_1^2[W]$ is well defined. Now define $\mu(\partial W, s)$ to be the residue class, modulo 1, of the rational number $(p_1^2[W] - 4\sigma)/896$. It is not difficult to show that this is an invariant of the spin manifold $(\partial W, s)$: that is, it does not depend on the choice of $(W, s')$.

*Example* 1. Let $M^7$ be a homotopy sphere which bounds a parallelizable manifold $W$ having signature $\sigma = +8$. (Compare Kervaire and Milnor [9].) Then

$$\mu(M^7) \equiv (0 - 4.8)/896 \equiv -\tfrac{1}{28} \pmod 1.$$

But according to [9] the group $\Theta_7$ consisting of all homotopy 7-spheres is cyclic of order 28, with generator $(M^7)$. It follows easily that $\mu$ provides a complete invariant for homotopy 7-spheres.

*Example* 2. The real projective space $P^{2n+1}$ can be thought of as the total space of a circle bundle over the complex projective space $P_n(C)$. (An explicit projection map is given by the formula,

$$(x_0 : y_0 : x_1 : y_1 : \cdots : x_n : y_n) \to (x_0 + iy_0 : x_1 + iy_1 : \cdots : x_n + iy_n).)$$

Let $\xi$ denote the associated 2-disk bundle, and let $W$ denote its total space, so that $\partial W = P^{2n+1}$. If $n$ is odd then $P_n(C)$ is a spin manifold; and it follows that $W$ is a spin manifold. The Euler class $e(\xi)$ is equal to twice a generator:

$$e(\xi) = 2a \quad \in \quad H^2(P_n(C); Z),$$

hence the Pontrjagin class $p_1(\xi) = e^2$ is equal to $4a^2$. The first Pontrjagin class for the tangent bundle of $P_n(C)$ is equal to $(n+1)a^2$. Therefore the class

$$p_1(W) \quad \in \quad H^4(W; Z) \cong H^4(P_n(C); Z)$$

corresponds to $(n+5)a^2$.

Now let us specialize to the case $n = 3$, and compute $p_1^2[W]$. From the sequence

$$0 \to H^4(W, P^7; Z) \to H^4(W : Z) \to H^4(P^7; Z) \to 0,$$

where the first two groups are infinite cyclic and the third is $Z_2$, one sees that the class $p_1(W) = 8a^2$ lifts back to 4 times a generator. By Poincaré duality the groups $H^4(W, P^7; Z)$ and $H^4(W; Z)$ are dually paired to $H^8(W, P^7; Z)$ by the cup product. Therefore $p^2[W] = \pm 4.8$. Choosing the orientation of $W$ correctly, and letting $s_1$ denote the induced spin structure on $\partial W = P^7$, this yields

$$\mu(P^7, s_1) \equiv (p_1^2[W] - 4\sigma)/896 \equiv (32 - 4)/896 \equiv \tfrac{1}{32} \pmod 1.$$

Thus we have computed the invariant $\mu$ for $P^7$ with one of its two possible spin structures.

Now consider the manifold $-W$ with reversed orientation. Its boundary is a spin manifold $(-P^7, s')$ with $\mu(-P^7, s') \equiv -\tfrac{1}{32}$. But the projective space $P^7$ possesses an orientation reversing diffeomorphism. This proves that there exists a spin structure $s_2$ on $P^7$ so that $\mu(P^7, s_2) \equiv -\tfrac{1}{32}$. In other words:

LEMMA 4. *The invariant $\mu$ of $P^7$ is equal to $\pm\tfrac{1}{32}$, where the sign depends on which of the two possible spin structures is chosen.*

Now consider connected sums of the form $P^7 \# M^7 \# \cdots \# M^7$; where $M^7$ is the homotopy sphere considered earlier. It can be shown that each such sum is homeomorphic to $P^7$. Clearly

$$\mu(P^7 \# M^7 \# \cdots \# M^7) \equiv \pm\tfrac{1}{32} - \tfrac{1}{28} - \cdots - \tfrac{1}{28}.$$

Since one can obtain 28 distinct invariants in this way, we have proved that the oriented manifold $P^7$ admits 28 distinct differentiable structures.

Let $r(M^7)$ denote the sum of $r$ copies of $M^7$. Next note that the 2-fold covering space of $P^7 \# r(M^7)$ is diffeomorphic to $S^7 \# 2r(M^7)$. Taking

$r = 14$ it follows that the 2-fold covering space of $P^7 \# 14(M^7)$ is diffeomorphic to the standard sphere $S^7$. In other words there exists an involution of $S^7$ whose orbit space is diffeomorphic to $P^7 \# 14(M^7)$. Since $P^7 \# 14(M^7)$ is distinct from $P^7$; this proves Theorem 1.

### 3. Spin cobordism groups

Using Lemma 2 one sees that the relation of "spin cobordism" between closed $n$-dimensional spin manifolds is an equivalence relation. Let $\Omega_n^{\text{spin}}$ denote the resulting cobordism group. The cartesian product operation between spin manifolds (compare Lemma 3) gives rise to a product operation

$$\Omega_m^{\text{spin}} \otimes \Omega_n^{\text{spin}} \to \Omega_{m+n}^{\text{spin}}.$$

Thus one obtains a graded ring.

Our ignorance concerning these groups is quite extensive. For example, to the best of my knowledge, it is not known whether any $\Omega_n^{\text{spin}}$ contains an element of order 4.

In studying spin cobordism it is natural to look at the homomorphism $\Omega_n^{\text{spin}} \to \Omega_n$ into the ordinary (oriented) cobordism group. Here is a description of this homomorphism for three particular cases.

For $n = 8$ the group $\Omega_8^{\text{spin}} \cong Z + Z$ maps monomorphically into $\Omega_8$, which is also isomorphic to $Z + Z$. The quotient group $\Omega_8/\Omega_8^{\text{spin}}$ is cyclic of order $2^7$. (Compare [11].)

Now let $n = 9$. According to Adachi [1] an oriented 9-manifold bounds if and only if the Stiefel–Whitney numbers $w_2^3 w_3[M]$ and $w_2 w_7[M]$ are zero. (Compare Wall [12].) But these clearly vanish for a spin manifold. Therefore the homomorphism

$$\Omega_9^{\text{spin}} \to \Omega_9 \cong Z_2 + Z_2$$

is zero. (Similarly this homomorphism is zero for $n = 5$, 11, and presumably for many higher dimensions.) On the other hand we will see presently that $\Omega_9^{\text{spin}} \neq 0$.

Finally, let $n = 10$. Let $H_{2,4}$ denote a non-singular hypersurface of degree (1, 1) in the product $P_2(C) \times P_4(C)$ of complex projective spaces. Then one can verify that

$$w_2(H_{2,4}) = 0, \qquad w_4 w_6[H_{2,4}] \neq 0.$$

Thus $H_{2,4}$ is a spin manifold, but is not an oriented boundary. It follows that the homomorphism

$$\Omega_{10}^{\text{spin}} \to \Omega_{10} \cong Z_2$$

is non-zero.

An extremely interesting tool for studying the spin cobordism ring was suggested to the author by Atiyah. Namely there is a homomorphism

$$\alpha\colon \Omega_n^{\text{spin}} \to \tilde{K}_O S^n$$

which is defined as follows. (Here $\tilde{K}_O S^n$ denotes the group of $s$-classes of vector bundles over $S^n$.) Consider a disk-bundle with spin structure, the fibre dimension being $8k$. Let $E$ denote the total space, $\dot{E}$ the total space of the associated sphere bundle, and $E/\dot{E}$ the Thom space. The two half-spin representations of Spin $(8k)$ give rise to two vector bundles over $E$. Since these representations coincide over Spin $(8k-1)$, the two vector bundles coincide over $\dot{E}$. Thus their difference determines an element of $K_O(E, \dot{E}) \cong \tilde{K}_O(E/\dot{E})$.

Applying this construction to the universal Spin $(8k)$-bundle, with $k$ large we obtain a canonical class $u \in \tilde{K}_O M$ (Spin $8k$). Now for any map $f\colon S^{n+8k} \to M(\text{Spin } 8k)$ the class $f^*u \in \tilde{K}_O S^{n+8k} \cong \tilde{K}_O S^n$ is defined. Thus we obtain a homomorphism from

$$\pi_{n+8k} M(\text{Spin } 8k) \cong \Omega_n^{\text{spin}}$$

to $\tilde{K}_O S^n$.

For $n \leq 7$ one can verify that this construction defines an isomorphism between $\Omega_n^{\text{spin}}$ and $\tilde{K}_O S^n$. (Compare [11].)

If $n \equiv 0 \pmod 4$ then the composition

$$\Omega_n^{\text{spin}} \xrightarrow{\alpha} \tilde{K}_O S^n \subset \tilde{K}_U S^n \cong Z$$

turns out to be precisely the $\hat{A}$-genus. In fact this is essentially the method which Atiyah and Hirzebruch [3] used to prove that the $\hat{A}$-genus of a spin manifold is an integer (or an even integer for $n \equiv 4 \pmod 8$).

Now recall that the tensor product operation between vector bundles gives rise to a bilinear product operation $\tilde{K}_O S^m \otimes \tilde{K}_O S^n \to \tilde{K}_O S^{m+n}$. (See [5].) The following result will be proved in a forthcoming paper by Atiyah, Bott, and Shapiro [2].

THEOREM. *The function* $\alpha\colon \Omega_n^{\text{spin}} \to \tilde{K}_O S^n$ *is a ring homomorphism: that is* $\alpha(M \times N) = \alpha(M)\alpha(N)$.

One interesting consequence of this theorem is the following. Let $n = 9, 10, 17$, or $18$.

THEOREM 2. *For these n there exists a smooth manifold* $\Sigma^n$ *which is a homotopy n-sphere, but is not a spin boundary.*

Thus the homomorphism $\Omega_n^{\text{spin}} \to \Omega_n$ has a non-trivial kernel for $n = 9, 10, 17, 18$.

PROOF. First consider the case $n \equiv 1 \pmod 8$. Let $k = (n-1)/4$.

According to Kervaire and Milnor [8] there exists an almost parallelizable manifold $M^{4k}$ with

$$\hat{A}[M^{4k}] = B_k a_k j_k/4k,$$

where $B_k$ is the $k$th Bernoulli number, $a_k = 1$ for $k$ even, and where $j_2 = 240$, $j_4 = 480$. Thus for $k = 2, 4$ there exists such a manifold with $\hat{A}[M^{4k}] = 1$. We may assume that $M^{4k}$ is $(2k - 1)$-connected, hence has a unique spin structure.

Let $(S^1, s)$ denote the circle with the "bad" spin structure, so that $\alpha(S^1, s) \neq 0$. Since the product

$$\tilde{K}_O S^{4k} \otimes \tilde{K}_O S^1 \to \tilde{K}_O S^{4k+1}$$

is non-trivial for $k$ even, it follows that $\alpha(M^{4k} \times (S^1, s)) \neq 0$. Now apply a surgery to $M^{4k} \times S^1$ so as to kill the infinite cyclic fundamental group. care must be taken that the surgery is compatible with the spin structure. (Compare [10, Theorem 2] or [9, §6].) The resulting manifold will be a $(2k - 1)$-connected spin manifold which is parallelizable except along an imbedded circle. Now apply a sequence of surgeries to kill the free abelian group $\pi_{2k}$. The resulting manifold $\Sigma^{4k+1}$ will be a homotopy sphere which is spin-cobordant to $M^{4k} \times (S^1, s)$; so that $\alpha(\Sigma^{4k+1}) \neq 0$.

To obtain a corresponding example in dimension $4k + 2$ it is only necessary to apply a single surgery to $\Sigma^{4k+1} \times (S^1, s)$. Again the surgery must be compatible with the spin structure. This completes the proof of Theorem 2.

It may be conjectured that similar manifolds $\Sigma^n$ exist whenever $n \equiv 1$ or $2 \pmod 8$. The above proof works for $n = 4k + 1$ whenever $B_k j_k/4k \equiv 1 \pmod 2$. An alternative attack would be based on the following.

CONJECTURE. *For $n \equiv 1 \pmod 8$ there exists a map $f\colon S^{8r+n} \to S^{8r}$ so that the induced $f^*\colon \tilde{K}_O S^{8r} \to \tilde{K}_O S^{8r+n} \cong Z_2$ is non-zero.*

(The corresponding statement for $n \equiv 2 \pmod 8$ would be a consequence.)

This is related to Theorem 2 as follows. Consider the composition

$$S^{8r} \subset M(\text{Spin } 8r) \xrightarrow{u'} BO$$

where $u'$ corresponds to the Atiyah–Hirzebruch class $u \in K_O M(\text{Spin } 8r)$ which was described above. Thus any element of $\pi_{8r+n}(S^{8r})$ determines a spin cobordism class. If $n \not\equiv 2 \pmod 4$ then it follows from [9] that this cobordism class has a representative which is a homotopy sphere. Now, assuming the Conjecture, it is not difficult to prove the generalized Theorem 2.

Conversely, using Theorem 2, it is easy to verify that the Conjecture is true for $n = 9, 17$.

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